ISOM 3360 – Data Work Log & Next Steps

Nov 4, 2026 · @Ryan

Summary

The data side of the project is done. I built a training set of 6,117 days (1 Jan 2010 to 30 Sep 2026) and collected hourly Polymarket prices for all 234 Hong Kong temperature market days so far (13 Mar to 6 Nov 2026).

Three findings change the plan:

- **Brackets cut off the decimal.** The 31°C bracket pays out for 31.0–31.9°C.
- **Fees are smaller than we assumed.** The plan used 5 points; the real maximum is 1.25 points.
- **The benchmark is harder than we assumed.** At 8pm the evening before, the market's implied forecast misses by about 1.06°C (RMSE). That's the number to beat, not "same as yesterday" (2.07°C).

Deliverables

File	What it holds	Use it for
HKO_Training_Set_2010_2026.xlsx	6,117 rows × 33 inputs + target, raw observations, data profile, data dictionary, sources	Modelling; the Data Understanding section of the report
hko_training_set.csv	Same training table as a flat file	`pd.read_csv` in the notebook
hko_build_training_set.py	Downloads the HKO files and rebuilds both tables	Reproducibility; report appendix
Polymarket_HK_Temp_Markets.xlsx	2,603 bracket markets, ~200k hourly prices, summary per day, HKO Daily Extract, overview of findings	Benchmark, trading backtest, settlement checks

File	What it holds	Use it for
polymarket_hk_temp_scraper.py	Re-pulls the market list, hourly prices and Daily Extract	Refreshing the data before the final report
pm_oct13_probability_path.png	Hourly prices for the 13 Oct market	A figure for the progress report
HKO_Weather_Sep2026.xlsx	One-month sample (Sep 2026)	Already covered by the full set; can be dropped

Data sources

All data comes from public APIs, so there was no screen-scraping.

Source	What we took	Coverage	How
HKO climate CSVs	Daily max/min/mean temperature, humidity, dew point, pressure, rainfall, cloud (HKO HQ); sunshine, solar radiation, evaporation, wind (King's Park)	Decades of history; we use Dec 2009 to 30 Sep 2026	One file per variable from `data.weather.gov.hk/cis/csvfile/<station>/ALL/`
HKO Daily Extract	"Absolute Daily Max", the value Polymarket settles on	Mar to early Nov 2026	Monthly files `dailyExtract_YYYYMM.xml`
Polymarket Gamma API	Market list, brackets, rules, fees, results (series 11312)	13 Mar to 6 Nov 2026	`gamma-api.polymarket.com/events?series_id=11312`
Polymarket price history (CLOB API)	Hourly Yes price for every bracket	Same	`clob.polymarket.com/prices-history`, fidelity = 60 min

What we left out on purpose: the HKO 9-day forecast (the team chose observations only) and hourly HKO observations. Archived hourly snapshots only start mid-2020; see Next steps.

Findings that change the plan

Topic	What we found	What changes
Settlement station	Markets settle on the HKO Daily Extract "Absolute Daily Max" at HKO HQ (Tsim Sha Tsui). That's one station, not the max across HK. Our training target matches it on all 213 overlapping days.	Nothing. We're already training on the settlement number. If you watch MyObservatory, make sure it shows the HKO reading, not your nearest station.
Bracket rule	Brackets cut off the decimal: 31.0–31.9 → "31°C". All 107 days with a .5–.9 decimal settled to the lower bracket.	The classes are floor(max temp). Turn predicted temperatures into bracket probabilities using that rule.
Fees	Taker fee = shares × 0.05 × p × (1 − p), so at most 1.25¢ per share (at p = 0.50). Makers pay nothing and get a 25% rebate. Mar–Apr markets had no fees. (docs)	Replace the plan's flat "5 points" with this formula in the trading rule.
Benchmark	At 8pm the evening before, the market misses by 0.82°C MAE / 1.06°C RMSE and puts 30.5% on the bracket that wins. "Same as yesterday" misses by 1.73°C RMSE on the same days.	The model has to beat the market, not just persistence.
Intraday	The market sharpens as the day goes on: 1.06°C RMSE the evening before, 0.82°C at noon, 0.51°C at 3pm.	Most of the edge, if there is any, is the evening before or the morning of.
Data lag	HKO publishes its checked daily files about once a month (October isn't out as of 4 Nov).	Fine for training. Live trading would need yesterday's values from the real-time feeds.
Bad market days	13 & 14 Mar had no trading and settled to the lowest bracket by default. Two duplicate "arch-" events (18 & 19 May) settled to the wrong brackets.	Leave all four out of the backtest.

Data profile highlights

The target (HKO daily max, 2010–2026) averages 26.65°C, standard deviation 5.39°C, range 7.1–36.9°C, skewness −0.50. The 36.9°C on 9 Aug 2026 is HKO's new all-time record.

- **Instances and attributes:** 6,117 days (one row per day); 33 inputs in five groups: past weather (t-1, t-2, t-7), moving averages and day-to-day changes, sun and wind, time of year (sin/cos), and extra t-1 readings.
- **Missing values:** only in King's Park evaporation (64 days), wind direction (53) and wind speed (39). Each gap is filled with the last reading before it, which only looks backwards, so nothing from the future leaks in. The training set has no blanks.
- **Skewed variable:** rainfall (skewness 7.3, max 425 mm in a day, 122 days more than 3 standard deviations out). Try log(1 + rain) for linear models.
- **Wind direction:** it goes round in a circle (359° sits next to 1°), so it's coded as sin/cos rather than raw degrees.
- **Leakage check:** on sample dates, the t-1, t-7 and 7-day average values matched the raw observations for the right earlier days.
- **Trial 1 baseline (persistence):** predicting today's max = yesterday's max gives RMSE 2.07°C, MAE 1.55°C over 2010–2026.

Next steps

The progress materials are due `Nov 5, 2026` 11:59pm, the feedback meeting is `Nov 10, 2026`, and the final report is due `Nov 26, 2026`.

For the progress report (tomorrow)

- [ ] Paste the Data Understanding numbers above and the Data Profile table into the report
- [ ] Add the 13 Oct price chart and the market-vs-persistence comparison
- [ ] Write down the plan changes: bracket rule, fee formula, market as the benchmark

Modelling (before the feedback meeting)

- [ ] Agree a time-ordered split: train 2010–2022, validate 2023, test 2024 onwards (or walk-forward). No random K-fold.
- [ ] Trial 2: Ridge/Lasso on temperature + logistic regression on "max ⩾ cutoff"; scale the inputs first
- [ ] Trial 3: a pruned decision tree (CART); tune max_depth and min_samples_leaf on the validation year
- [ ] Trial 4: gradient boosting (XGBoost or sklearn's HistGradientBoosting); tune with time-series CV

- [ ] Turn temperature predictions into bracket probabilities: predicted mean + residual spread → P(floor(max) = k) for each bracket

Evaluation and backtest (before the final report)

- [ ] Score each model on RMSE/MAE (temperature) and Brier/log-loss (brackets); check calibration with reliability plots
- [ ] Compare against the market at 8pm on t-1 on the same days (p(winner), Brier, RMSE)
- [ ] Backtest the trading rule on Mar–Nov 2026 using the 8pm t-1 price, the real fee formula and a minimum edge; report net P&L and number of trades
- [ ] Re-run both scrapers just before submitting so October and November are included

Stretch (future work)

- [ ] Morning model: add hourly HKO observations (archived from mid-2020) to predict at 8am on the day
- [ ] Add the HKO 9-day forecast as an input (archived from Apr 2021) and see whether it beats the market

Open questions

- **Classification or regression?** Predicting the bracket directly (11 classes, so class imbalance in the tails) or predicting temperature and converting it to bracket probabilities. Our suggestion is to do both and compare.
- **When do we "trade"?** The backtest needs one fixed time. 8pm on t-1 matches what the model knows, but 8am on the day has more trading and a market that's harder to beat.
- **Market prices as inputs?** The plan keeps them out so the model stays independent of the market. Worth asking at the feedback meeting whether the instructor expects them as attributes, since the original brief listed them.
- **Is the backtest window long enough?** About 230 market days, so a profit figure will be noisy. Report it with a confidence interval or by month, not as one headline number.
- **Access:** we haven't checked whether placing real trades on Polymarket is allowed from Hong Kong. The project only needs a paper backtest, so that's enough.

Sources: HKO Open Data · HKO Climatological Information · Polymarket market rules (13 Oct example) · Polymarket fees